

AI Development Expert – Final Project

Business Plan

NegoPlay

Strategic Profiles from Bridge to Business:

A B2B AI-Powered Negotiation Intelligence Platform

Tagline: *“Know your counterpart before you walk into the room.”*

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1 Executive Summary

The Business in Three Sentences

NegoPlay is a B2B SaaS platform that automatically discovers decision-making profiles from elite competitive data and deploys them as AI agents to help professionals prepare for, simulate, and win business negotiations.

The core insight: 149,208 expert bridge hands reveal stable behavioral archetypes (“Slam Hunter”, “Insurance Player”, etc.) that transfer measurably to negotiation behavior. We turn this empirical finding into a subscription API for sales teams, procurement departments, and M&A advisors.

Target revenue in Year 3: \$480K ARR. **Total budget for MVP:** under \$15 (LLM costs only, excluding developer time).

Table 1.1: Business plan summary.

Category	Value
Market (TAM)	\$4.2B (negotiation intelligence + sales AI)
Market (SAM)	\$420M (mid-market + enterprise, English-speaking)
Market (SOM Y3)	\$480K ARR (realistic 5-year ramp)
Business model	B2B SaaS subscription + usage-based API
Primary customer	Sales directors, procurement managers, M&A teams
Pricing (entry)	\$49 / month per seat
Competitive moat	Empirically validated profiles from real expert data
Technology stack	Python, Gemini Flash 2.0, scikit-learn, GitHub

2 The Problem

2.1 Negotiations fail because we do not know our counterpart

Every business negotiation involves incomplete information. We know our own BATNA (Best Alternative to a Negotiated Agreement) but rarely know the other side's risk tolerance, decision style, or likely concession pattern. The result:

- Deals are left on the table when one side concedes too quickly.
- Deals collapse when one side misjudges the other's walk-away point.
- Sales cycles are prolonged because reps cannot adapt their style to the buyer's decision profile.

2.2 The data gap

Business negotiation lacks the clean feedback loops that make other domains trainable:

- Real negotiation transcripts are proprietary.
- Outcomes ("who won?") are rarely measurable in standardized form.
- No public dataset allows training behavioral models on real expert negotiation decisions.

Existing tools (Gong, Chorus, Clari) analyze *what was said* after the fact. No tool predicts *how the counterpart will behave* before the conversation starts.

The gap we fill

NegoPlay is the first platform to build behavioral profiles from a large, public, measurable expert decision dataset – bridge tournament records – and transfer those profiles to negotiation preparation.

3 The Solution

3.1 What NegoPlay does

NegoPlay discovers four to five decision-making archetypes from 149,208 elite bridge tournament hands. Each archetype is translated into a named profile with five to seven behavioral traits. The profiles are then deployed as AI agents that can:

1. **Profile your counterpart** – given publicly available information about the person or company, classify them into the nearest archetype.
2. **Simulate the negotiation** – run a multi-round AI-vs-AI simulation of the upcoming deal so the user can see likely concession patterns before the real meeting.
3. **Advise in real time** – via API integration with CRM tools (HubSpot, Salesforce), push profile-based suggestions during a live call.

3.2 The four profiles (confirmed by Stage 1)

Table 3.1: Decision profiles discovered from 149,208 bridge tournament hands. Profile names and assignment rates are the actual Stage 1 outputs (v2.1, May 2026).

Profile	n (%)	Core traits (from bridge data)	Predicted negotiation behavior
Slam Hunter	20 (3.6 %)	Bids slams at $1.84\times$ the population baseline; calculated upside bets; trusts partner’s strength signals.	Opens high, concedes little, walks away fast if the value gap is large.
Insurance Player	21 (3.7 %)	Stays in partscore at $1.20\times$ baseline; risk-averse; values certainty over upside.	Prefers small guaranteed gains; long close cycles; many contingencies in the contract.
NT Specialist	17 (3.0 %)	Plays NoTrump contracts at $1.36\times$ baseline; analytical; precise trick counting.	Data-driven; demands DCF / comparables justification; rejects “gut-feel” offers.
Fighter	37 (6.6 %)	Doubles opponents for penalties at $1.55\times$ baseline; competitive; aggressive against opponents.	Counter-offers aggressively; threatens walk-away as a tactic; rejects first offer reflexively.
Generalist (baseline)	468 (83.1 %)	Average rates on all axes; control population.	Default-reasonable; matches counterpart’s style.

Note on naming consistency. Earlier internal drafts of this plan used the working names “Bluffer” and “Doubler” for two of the profiles. After Stage 1 ran on the full 149K-row dataset, the actual statistical axes turned out to be `nt_rate` and `penalty_double_rate`, leading to the more descriptive names *NT Specialist* and *Fighter*. The Preliminary Report (Table A.2) and this Business Plan are now synchronised on these names.

3.3 Why bridge data works

Bridge is one of the only public, large-scale, measurable records of expert decision-making under incomplete information with four-player dynamics (two cooperating, two competing – exactly like a business negotiation involving two companies each with internal stakeholders). The IMP scoring system provides a clean, standardized outcome signal. No comparable dataset exists for business negotiation.

4 Market Analysis

4.1 Total Addressable Market (TAM)

The global sales intelligence software market was valued at approximately \$3.4B in 2023 and is growing at 11 % CAGR. Adjacent markets include procurement analytics (\$1.2B) and M&A advisory tools (\$0.6B). We define NegoPlay’s TAM as the intersection of these segments:

Table 4.1: TAM breakdown.

Segment	Market size (2024)	CAGR
Sales intelligence (enterprise)	\$2.8B	11 %
Procurement AI tools	\$0.9B	15 %
M&A deal intelligence	\$0.5B	9 %
Total TAM	\$4.2B	11 %

4.2 Serviceable Addressable Market (SAM)

NegoPlay targets mid-market and enterprise companies in English-speaking markets (US, UK, Australia, Canada) that: (a) use a CRM, (b) have active outbound sales or procurement teams, and (c) have an existing AI software budget. This narrows the addressable market to approximately 10 % of the TAM.

Table 4.2: SAM calculation.

Parameter	Value
TAM	\$4.2B
English-speaking market share	× 40 %
Mid-market + enterprise filter	× 60 %
AI-ready companies	× 45 %
SAM	≈ \$450M

4.3 Serviceable Obtainable Market (SOM)

In the first three years, NegoPlay targets 0.1 % of the SAM through direct sales to early adopters (academic networks, bridge-community connections, AI-forward SMEs).

Table 4.3: SOM projection over three years.

Metric	Year 1	Year 2	Year 3
Paying customers (seats)	20	120	800
Average revenue per seat/month	\$49	\$55	\$50
Monthly Recurring Revenue	\$980	\$6,600	\$40,000
Annual Recurring Revenue (ARR)	\$11,760	\$79,200	\$480,000

5 Business Model and Pricing

5.1 Revenue model

NegoPlay uses a **freemium-to-subscription** model with an optional usage-based API tier for enterprise integrations.

Table 5.1: Pricing tiers.

Tier	Audience	Includes	Price
Free	Individual / researcher	10 profile analyses/month, no simulation	\$0
Pro	Sales rep / procurement officer	200 analyses + unlimited simulations + email support	\$49/seat/month
Team	SME sales team (up to 20 seats)	Unlimited analyses, CRM web-hook, team dashboard	\$199/month flat
Enterprise	Large enterprise / API integration	Unlimited, SLA, SSO, Salesforce / HubSpot connector	Custom (est. \$2,000-\$10,000/month)

5.2 LLM pricing model – input vs. output tokens

LLM providers price input tokens (the prompt the model reads) and output tokens (the response the model generates) at *different* rates. Output tokens are typically 4× more expensive because generation requires more compute per token than reading. Any honest cost model must therefore track both directions separately.

Table 5.2: Per-million-token pricing across LLM providers (May 2026). Output is consistently more expensive than input.

Provider / model	Input	Output	Out/In ratio
Gemini Flash 2.0 (default)	\$0.075 / M	\$0.30 / M	4.0×
Gemini 2.5 Pro	\$1.25 / M	\$5.00 / M	4.0×
Claude Haiku 4.5	\$1.00 / M	\$5.00 / M	5.0×
Claude Opus 4.7	\$15.00 / M	\$75.00 / M	5.0×
GPT-4o	\$2.50 / M	\$10.00 / M	4.0×
GPT-4o mini	\$0.15 / M	\$0.60 / M	4.0×

The cost of a single LLM call therefore decomposes as

$$C_{\text{call}} = \frac{T_{\text{in}}}{10^6} \cdot c_{\text{in}} + \frac{T_{\text{out}}}{10^6} \cdot c_{\text{out}} \quad (5.1)$$

where $T_{\text{in}}, T_{\text{out}}$ are the input and output token counts and $c_{\text{in}}, c_{\text{out}}$ are the per-million-token unit prices.

5.3 Unit economics per analysis

The core cost driver is LLM token consumption. Each NegoPlay profile analysis sends a long prompt (system instruction + counterpart context) and receives a relatively short structured response. The input/output breakdown matters because input dominates the token *count* but output dominates the token *cost*.

Table 5.3: Unit economics per profile analysis, with input and output costs separated. Note that 38 % of total tokens are output, but they account for 71 % of the LLM cost.

Component	Tokens	Unit rate	Cost
Gemini Flash 2.0 – input (system + user)	3,000	\$0.075 / M	\$0.000225
Gemini Flash 2.0 – output (structured JSON)	1,800	\$0.300 / M	\$0.000540
LLM subtotal per analysis	4,800	—	\$0.000765
Infrastructure (hosting, storage, logging)	—	—	\$0.005
Total cost per analysis	—	—	\$0.0058
Revenue per analysis (Pro)	—	—	\$0.245 (200 calls / \$49)
Gross margin per analysis	—	—	97.6 %

5.4 Token economics summary across pipeline stages

Table 5.4: Token economics across the four pipeline stages, with input and output costs separated. The Stage 2 skill-extraction call is the most expensive single call because the LLM reads many bridge hands (high T_{in}); per-turn negotiation calls are dominated by output cost.

Stage	T_{in}	T_{out}	In cost	Out cost	Total
Stage 2 – Skill extraction	7,000	1,000	\$0.000525	\$0.000300	\$0.000825
Stage 3 – Agent prompt assembly	1,600	400	\$0.000120	\$0.000120	\$0.000240
Stage 4 – Bridge bidding turn	2,500	500	\$0.000188	\$0.000150	\$0.000338
Stage 4 – Negotiation turn	3,500	1,500	\$0.000263	\$0.000450	\$0.000713
Full pipeline (one user analysis)	14,600	3,400	\$0.001096	\$0.001020	\$0.002116

Why this split matters commercially

18 % of NegoPlay’s tokens are output, but they account for 48 % of the LLM cost. Output prompt design is therefore a direct lever on gross margin – returning concise structured JSON instead of verbose free text cuts the per-call cost roughly in half. Our agent prompts (see Listing in the Preliminary Report) require strict JSON schemas of at most 1,500 output tokens for this reason.

Cost advantage of Gemini Flash 2.0

With **input at \$0.075 / M** and **output at \$0.30 / M**, Gemini Flash 2.0 is approximately 10× cheaper than GPT-4o (\$2.50 / \$10.00) for equivalent throughput, and roughly 200× cheaper than Claude Opus 4.7 (\$15 / \$75). The entire NegoPlay MVP (development + testing) runs for under \$15 in LLM costs. Claude and GPT are used selectively for cross-model validation runs to strengthen publishability, not for production traffic.

6 Competitive Landscape

6.1 Direct and indirect competitors

Table 6.1: Competitive landscape.

Competitor	What they do	Gap vs NegoPlay	Threat
Gong.io	Analyzes past sales calls (voice + transcript) to coach reps.	Post-hoc only; no pre-meeting counterpart profiling; no game-theory basis.	Medium
Pactum AI	Autonomous negotiation bot for procurement (IKEA, Walmart).	Fully automated (no human in the loop); no behavioral profiling of individuals.	Low–Medium
Chorus.ai	Records and transcribes sales calls; CI insights for managers.	Same as Gong; reactive, not predictive.	Low
Clari	Revenue forecasting + deal risk scoring.	Forecasting only; no behavioral simulation.	Low
Crystal Knows	DISC personality profiling from LinkedIn data.	Not empirically validated on expert decision data; no simulation.	Medium
NegoPlay	Profiles + simulates counterpart from empirically validated archetypes.	<i>Only platform with an empirical game-theoretic foundation.</i>	–

6.2 Our competitive moat

1. **Empirical foundation:** Profiles are derived from 149,208 expert decisions with measurable outcomes – not from personality questionnaires or self-reported surveys. This is defensible in peer-reviewed literature.
2. **Cross-domain validation:** The alignment finding (Spearman $\rho \geq 0.7$ between bridge and negotiation behavior) can be published – creating an academic moat.
3. **Provider-neutral architecture:** The unified `llm_client.py` wrapper means NegoPlay is not locked to any single LLM provider.
4. **Data accumulation:** Every new customer interaction refines the profiles – a classic data flywheel.

7 Go-to-Market Strategy

7.1 Phases

Phase 1 – Academic validation (Months 1–6, free)

- Publish the alignment finding as a conference paper (AAMAS or IJCAI).
- Open-source the profile discovery code on GitHub to build credibility.
- Target bridge clubs, negotiation researchers, and MBA programs as early adopters and free validators.
- **Goal:** 100 registered free users, 1 academic citation.

Phase 2 – Paid early access (Months 7–18)

- Launch Pro tier (\$49/seat/month) via Product Hunt and LinkedIn.
- Leverage the author’s networks at Sami Shamoon College and LUT University.
- Offer a 3-month pilot to 3–5 mid-market sales teams at cost.
- **Goal:** 20 paying customers, \$11,760 ARR.

Phase 3 – Enterprise and API (Months 19–36)

- Build HubSpot and Salesforce connectors.
- Pursue enterprise deals with procurement departments (indirect sales via consultancies).
- Explore white-labeling for negotiation training companies.
- **Goal:** 800 seats, \$480K ARR.

7.2 Customer acquisition cost estimate

Given that Phase 1 is entirely inbound (open source + academic press), early customer acquisition cost (CAC) is near zero. In Phase 2, assuming \$2,000/month in LinkedIn advertising and content marketing:

- **CAC (Phase 2):** $\$2,000 / 20 \text{ customers} = \100 per customer .

- **LTV (Pro, 12-month retention):** $\$49 \times 12 = \588 .
- **LTV/CAC ratio:** $5.9 \rightarrow$ healthy (target > 3).

8 Roadmap and Milestones

Table 8.1: 18-month product and business roadmap.

Month	Milestone	Type
1–5	Complete NegoPlay MVP (4-stage pipeline, 4 agents, alignment result)	Technical
5	Submit alignment paper to AAMAS / AAAI workshop	Academic
6	Open-source profile-discovery code on GitHub	Marketing
7	Launch Pro tier on Product Hunt	Commercial
9	20 paying customers; first case study published	Commercial
12	HubSpot webhook integration (beta)	Technical
15	First enterprise pilot (procurement team)	Commercial
18	\$79K ARR; Salesforce connector GA	Commercial
24	Series A readiness assessment or academic spin-off decision	Strategic

9 Financial Projections

9.1 Three-year revenue projection

Table 9.1: Three-year revenue, cost, and margin projection.

Item	Year 1	Year 2	Year 3
<i>Revenue</i>			
Pro seats (avg)	20	120	600
Pro revenue	\$11,760	\$79,200	\$352,800
Enterprise / API revenue	—	—	\$127,200
Total ARR	\$11,760	\$79,200	\$480,000
<i>Costs</i>			
LLM API – input tokens (\$0.075/M)	\$182	\$1,221	\$7,403
LLM API – output tokens (\$0.300/M)	\$171	\$1,155	\$6,997
LLM API subtotal (Gemini, $\approx 97\%$ GM)	\$353	\$2,376	\$14,400
Infrastructure (cloud hosting)	\$1,200	\$3,600	\$12,000
Marketing (LinkedIn, content)	\$0	\$24,000	\$60,000
Total COGS + OpEx	\$1,553	\$29,976	\$86,400
Gross Profit	\$10,207	\$49,224	\$393,600
Gross Margin	87 %	62 %	82 %

9.2 Funding requirement

Phase 1 requires **zero external capital** (developer is the founder; LLM costs are under \$15 for the full MVP). Phase 2 marketing costs (\$24K/year) can be bootstrapped from Phase 1 revenue. A seed round (\$200K–\$500K) would be sought in Year 2 to fund enterprise sales and engineering headcount.

10 Business Risks

Table 10.1: Commercial risk register.

#	Risk	Lik.	Imp.	Mitigation
BR1	Research finding not supported ($\rho < 0.5$).	Med	High	Null result is publishable; pivot to “bridge coaching AI” (profile analytics without negotiation transfer claim).
BR2	Gong or Salesforce launches a competing feature.	Med	High	Academic moat (peer-reviewed publication) + niche positioning (empirical game-theory basis) differentiates us.
BR3	LLM providers raise prices significantly.	Low	Med	Provider-neutral <code>llm_client.py</code> lets us switch to the cheapest provider per task within days.
BR4	Data privacy regulations block use of player name data.	Low	Med	Player data is public (tournament websites); we profile <i>negotiation counterparts</i> , not users.
BR5	The bridge-to-negotiation transfer claim is rejected by peer reviewers.	Med	Med	Frame as “proof of concept in simulated environment” from the start; null result is reported honestly.
BR6	Slow enterprise sales cycle (>9 months).	High	Med	Focus on SME Pro tier first; enterprise is a Year 2+ play.

11 Team

Founder Profile: Anna Ben-Shushan

- **Current role:** Lecturer at Sami Shamoon College of Engineering; PhD candidate at LUT University (Finland).
- **Academic:** M.Sc. Software Engineering; AI Development Expert course (current).
- **Industry:** Full-stack developer; AI consultant.
- **PhD topic:** “From Bridge Tables to Global Markets: AI and Optimization Models of Cooperation, Competition, and Negotiation on Digital Platforms.”
- **Unique advantage:** Domain expert in both the data source (bridge) and the application domain (business negotiation), with the academic credibility to publish the alignment finding.

11.1 Hiring plan (if funded)

- **Month 9:** Part-time front-end developer (student, paid project).
- **Month 15:** Full-time ML engineer (seed-funded).
- **Month 18:** Business development manager for enterprise sales.

12 Conclusion

NegoPlay sits at the intersection of three trends: the rapid adoption of AI in enterprise software, the growing demand for data-driven negotiation tools, and the academic maturation of multi-agent LLM systems.

Its core differentiator is an **empirically grounded behavioral model** – profiles discovered from 149,208 expert decisions rather than assembled from questionnaires. This gives NegoPlay both a commercial moat and an academic story that can be defended in peer review.

The financial model is attractive: 97% gross margins on LLM calls, near-zero CAC in Phase 1, and a clear path to \$480K ARR by Year 3 without external capital for the first 18 months.

The single most important number

Spearman $\rho \geq 0.7$

If the cross-domain alignment exceeds this threshold, NegoPlay has a publishable result, a defensible product claim, and a credible reason for enterprise customers to pay.

Everything else follows from this number.